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PEPPANET: EFFECTIVE MISPRONUNCIATION DETECTION AND DIAGNOSIS LEVERAGING PHONETIC, PHONOLOGICAL, AND ACOUSTIC CUES

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ABSTRACT

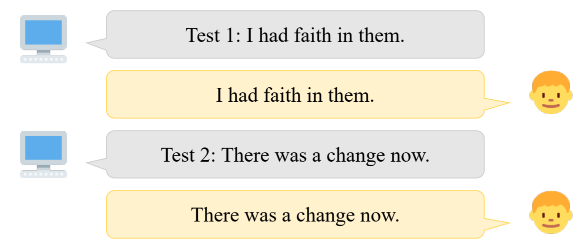
Mispronunciation detection and diagnosis (MDD) aims to detect erroneous pronunciation segments in an L2 learner's articulation and subsequently provide informative diagnostic feedback. Most existing neural methods follow a dictation-based modeling paradigm that finds out pronunciation errors and returns diagnostic feedback at the same time by aligning the recognized phone sequence uttered by an L2 learner to the corresponding canonical phone sequence of a given text prompt. However, the main downside of these methods is that the dictation process and alignment process are mostly made independent of each other. In view of this, we present a novel end-to-end neural method, dubbed PeppaNet, building on a unified structure that can jointly model the dictation process and the alignment process. The model of our method learns to directly predict the pronunciation correctness of each canonical phone of the text prompt and in turn provides its corresponding diagnostic feedback. In contrast to the conventional dictation-based methods that rely mainly on a free-phone recognition process, PeppaNet makes good use of an effective selective gating mechanism to simultaneously incorporate phonetic, phonological and acoustic cues to generate corrections that are more proper and phonetically related to the canonical pronunciations. Extensive sets of experiments conducted on the L2-ARCTIC benchmark dataset seem to show the merits of our proposed method in comparison to some recent top-of-the-line methods.

Index Terms— Computer-assisted pronunciation training (CAPT), mispronunciation detection and diagnosis (MDD), dictation model, text prompt, L2-ARCTIC

1. INTRODUCTION

Developments of computer-assisted pronunciation training (CAPT) systems open up new possibilities to practice L2 (second language) pronunciation skills in a stress-free and self-directed manner. CAPT systems also figure prominently in supplementing teachers' instructions, meanwhile alleviating the lack of qualified teachers [1][2]. In a typical CAPT system, L2 learners are presented with a text prompt and asked to read it aloud. According to the input speech, the system can inform learners about their mispronounced phone segments, and subsequently the learners can repeat reading-aloud training to steadily improve their pronunciation proficiency. As an indispensable ingredient of CAPT systems, mispronunciation detection and diagnosis (MDD) is expected to assess not only pronunciation proficiency at the supra-segmental level (e.g., prosody and intonation) [3][4][5], but also mispronunciations at the segmental level (e.g., phone and word) [6][7]. Since assessments of

Reading-aloud Training



Mispronunciation Detection and Diagnosis Results

MDD result of test 1

	Had			Faith			In		Them		
PMT	hh	ae	d	f	ey	th	ih	n	dh	eh	m
DIC	hh	ae	d	f	t	-	ih	n	d	unk	m
DIAG	hh	ae	d	f	t	sil	ih	n	d	unk	m
ES	0	0	0	0	1	1	0	0	1	1	0

MDD result of test 2

	There				Was			A	Change			
PMT	dh	eh		r	w	aa	z	ah	ch	ey	n	jh
DIC	z	eh	ah	unk	w	aa	z	ah	ch	ey	n	jh
DIAG	z	eh	-	unk	w	aa	z	ah	ch	ey	n	jh
ES	1	0	-	1	0	0	0	0	0	0	0	0

Fig. 1. Examples selected from the L2-ARCTIC dataset, which show the processing flow of reading-aloud training with phone-level mispronunciation detection and diagnosis. PMT denotes the corresponding canonical phone sequence of a text prompt; DIC refers to a dictated phone sequence generated by a dictation-based MDD model; DIAG indicates the manual annotation result; and ES indicates the mispronunciation detection result.

the former are even challenging for human to achieve high-level inter-rater agreement, most previous endeavors and applications have focused exclusively on the latter.

As such, there has been a surge of recent research interest in developing techniques to evaluate L2 learners' pronunciation quality and meanwhile pinpoint their phone-level mispronunciations. Among the traditional pipeline methods, pronunciation-scoring based ones are the most prominent and widely used. These methods, in general, compute a ratio between the likelihoods of a canonical phone and the most likely pronounced phones obtained by forced-aligning the canonical phone sequence of a text prompt to the learner's actual articulation. Accordingly, phone-level mispronunciations are then detected with phone-dependent

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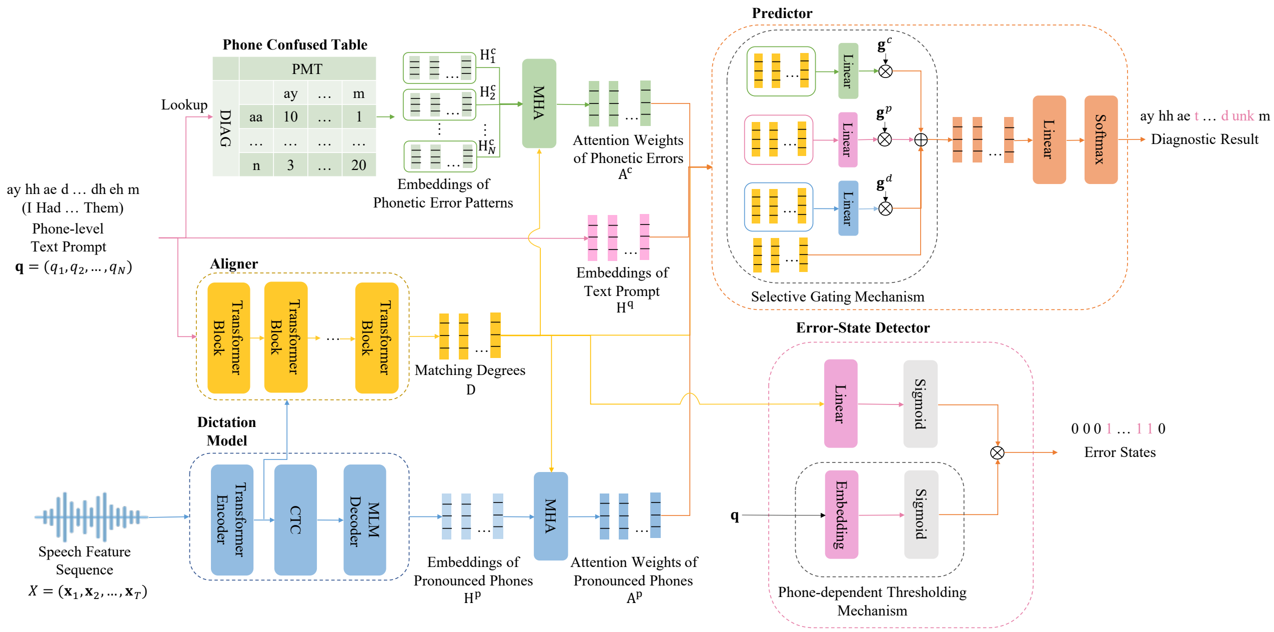

Fig. 2. A schematic depiction of the model structure of our MDD method.

thresholds [8][9][10]. Some other recent methods frame MDD as a dictation task by employing a free-phone recognition process to dictate the most likely phone sequence uttered by an L2 learner. In so doing, the erroneous pronunciation portions can be readily identified by comparing the dictation result with the corresponding canonical phone sequence [11]-[14]. Figure 1 illustrates a generic process of the dictation-based MDD paradigm; an L2 learner practices the pronunciations of "faith", "them" and "there." In response, the MDD system dictates the learner's phone-level pronunciations to be /f t/, /d unk m/ and /z eh ah unk/. Through aligning the pronounced phone sequences with the canonical phone sequence of a text prompt (/f ey th/, /dh eh m/ and /dh eh r/, respectively), the system can output the error state of each phone of the canonical phone sequence and provide the corresponding diagnostic feedback.

Albeit the effectiveness of the existing MDD methods, there remain two under-explored problems. On one hand, the dictation process and alignment process are usually made independent of each other, probably leading to unlikely inconsistency between the dictation result and the oracle diagnostic feedback. For example, in Figure 1, the oracle (human-annotated) diagnosis feedback of "faith" and "there" focuses exclusively on substitution and deletion errors while ignoring insertion errors that will also be indicated by the dictation model (the phonemes marked in blue). To provide the oracle diagnosis feedback, human assessors first take into account the canonical phone sequence of the text prompt and then compare it with the perceived learner's articulation to locate where phone-level pronunciation errors occur. On the other hand, most of the MDD models fall short in taking advantage of the prior dependencies of phonetic similarities between the actual pronounced phones of L2 learners and their corresponding canonical phones. Specifically, the MDD models output the most likely pronounced phone symbols via a dictation process, which solely depends on speech features and is prone to generate corrections with totally different phonetic attributes. The mispronounced phones and their corresponding canonical phones typically share some common phonetic attributes or similarities. In Figure 1, the canonical phones /dh/ and /iy/ are mispronounced as /d/, /z/ and /ih/ (marked in red), respectively. The phones sets {/dh/, /d/, /z/} and {/iy/, /ih/} can be characterized to have the dental and palatal attributes, respectively [15][16].

Building on these observations, we in this paper present a novel end-to-end neural MDD method, dubbed PeppaNet, which leverages a unified architecture to jointly model the dictation process and alignment process. PeppaNet capitalizes on the phonetic, phonological and acoustic cues gathered from frequently **P**ronunciation **E**rror patterns, **P**erceived **P**hone sequences, and learner's **A**rticulation traits. As such, PeppaNet can employ an error-state detector network to directly estimate the correctness of each phone token in the canonical phone sequence, and uses a predictor network to generate the corresponding diagnostic result. Instead of dictating the learner's articulation in a prompt-agnostic manner, the predictor network is equipped with a novel selective gating mechanism to generate corrections that are more proper and phonetically similar to canonical pronunciations. In summary, the contributions of this paper include: (1) we propose a unified neural model structure to tightly integrate the dictation process and the alignment process for better MDD performance; (2) we also design a novel selective gating mechanism to fuse disparate kinds of information cues; (3) extensive empirical experiments are conducted on a public benchmark dataset to evaluate of the feasibility of our proposed method; (4) our method is readily applicable to practical use cases wherein only a limited amount of annotated data is made available.

2. PROPOSED METHOD

2.1. Problem Definition

We tackle the mispronunciation detection and diagnosis (MDD) task with the following processing flow. Given a *T*-length speech feature sequence $X = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)$ extracted from an L2 learner's utterance and a given text prompt that contains a *N*-length canonical phone sequence $\mathbf{q} = (q_1, q_2, \ldots, q_N)$, an MDD model aims to detect error states $\mathbf{e} = (e_1, e_2, \ldots, e_N)$ in response to $\mathbf{q}$ and output the corresponding diagnostic feedback $\mathbf{y} = (y_1, y_2, \ldots, y_N)$, where y_n denotes the actual phone the learner has uttered with respect to q_n.

2.2. Model Components

The model architecture of proposed method is composed of five modules, as depicted in Figure 2, notably mentioned: 1) a lookup module that sequentially augments each phone in $\mathbf{q}$ with its most frequently mispronounced counterparts stored in the phone confusion table; 2) a dictation module that recognizes the most likely pronounced phone sequence according to X; 3) a soft aligner that calculates a sequence of matching degree vectors for each phone q_n of the text prompt $\mathbf{q}$; 4) a detector that takes input the matching degree vectors for each canonical phone q_n and estimates the probability that a pronunciation error has occurred with a corresponding phone-dependent threshold; and 5) a predictor that offers the diagnostic feedback $\mathbf{y}$ by taking input a fused representation generated from a novel selective gating mechanism.

2.3. Dictation Module

We adopt Mask-CTC as the workhorse of the dictation module [17][18], due to its fast inference speed and good recognition performance. Mask-CTC is a non-autoregressive neural model for speech recognition, which predicts a lexical or phonetic token sequence within a constant number of inference steps. Specifically, Mask-CTC follows an encoder-decoder neural model architecture, where CTC greedy decoding [19] is applied to the output vector sequence of the encoder to generate an initial recognition hypothesis at one step instead of incremental computation, which is subsequently optimized with the masked language model (MLM) objective function in an iterative manner [20].

In more precise terms, given an input speech feature X, a Transformer encoder first outputs a sequence of encoded acoustic feature vectors $\mathrm{H}^X = (\mathbf{h}_1^X, \mathbf{h}_2^X, \ldots, \mathbf{h}_{T'}^X)$. After performing CTC greedy decoding on the H^X, an initial hypothesis for the pronounced phone sequence of an L2 learner's utterance is generated by suppressing repeated tokens and removing blank symbols. A refined hypothesis $\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_L)$ is obtained by repeatedly predicting a masked subset of phones from the initial (or previous) hypothesis which have low confidence scores with an MLM decoder in a fixed number of k iterations.

2.3. Aligner

The aligner provides an explicit way to soft-align each phone q_n of the text prompt $\mathbf{q}$ with different encoded acoustic feature vectors $\mathbf{h}_t^X$ of H^X, determining whether q_n has really been uttered somewhere in the learner's utterance. We instantiate the aligner with a Transformer network that produces a sequence of matching degree vectors $\mathrm{D} = (\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_N)$ between $\mathbf{q}$ and H^X. Specifically, the aligner involves several transformer blocks, where each block consists of two multi-head attention (MHA) layers followed by a feed-forward network [21]. The first MHA layer in each block aims to capture the interactions between the corresponding phone embeddings $\mathrm{H}^{\mathrm{q}} = (\mathbf{h}_1^{\mathrm{q}}, \mathbf{h}_2^{\mathrm{q}}, \ldots, \mathbf{h}_N^{\mathrm{q}})$ of the text prompt $\mathbf{q}$. After that, the subsequent MHA layer produces a sequence of matching degree vectors D between the output of previous block and the encoded acoustic feature vectors by setting the key and value matrices both to be H^X.

2.5. Phone Confusion Table

To prevent our model from producing unreasonable MDD predictions given that the text prompt is known in advance, we leverage a phone confusion table to seize the dependence relationship of deviation (or substitution) patterns among phones for L2 learners' pronunciations. The confusion table can be inferred according to the statistics of incorrect (substituted) pronunciation patterns collected from the training set. To be specific, during the MDD process, we dynamically build a confusion table in the form of a matrix $\mathrm{M} \in \mathbb{R}^{V*N}$ for the whole phone sequence of a text prompt $\mathbf{q}$, where V is the number of phones in the inventory of canonical phones and N is length of $\mathbf{q}$. For a column n, each of its elements is the number of times that q_n is mispronounced as another phone in its corresponding canonical phone set, accumulated from the training set. To reduce the computational complexity, we merely select the top-K candidates for building the phone confusion matrix (i.e., $V = K$) with a sparse matrix scheme.

2.6. Error-State Detector

The error-state detector is a sequential binary labeling model where the core component is a phone-dependent thresholding mechanism. The matching degree matrix D is used in conjunction with $\boldsymbol{q}$ as the input for the detector. The output of the detector is a sequence of error states $\mathbf{e} = (e_1, e_2, \ldots, e_n)$, where e_n denotes the error state of the n-th phone in the text prompt: 1 indicates that phone q_n is incorrectly pronounced and 0 otherwise. To estimate the error states in a fine-grained manner for different canonical phones, we design a phone-dependent thresholding mechanism which makes use of a learning strategy to generate a specific threshold value $\mathrm{E}(q_n)$ for each canonical phone q_n. This mechanism can be implemented through an embedding layer followed by a sigmoid function. As such, the error states e_n of the n-th phone of $\mathbf{q}$ is formalized by

$$e_n = \sigma(\mathbf{w}_{\mathrm{e}}^T \mathbf{d}_n + \mathrm{b}^e) \times \mathrm{E}(q_n), \tag{1}$$

where $\mathrm{E}(\cdot)$ represents the output of our proposed phone-dependent thresholding mechanism, the vector $\mathbf{w}_{\boldsymbol{e}}$ and the scalar b^e are learnable parameters.

2.7. Predictor

The predictor performs a sequential multi-class labeling process, where the proposed selective gating mechanism serves as a gate to elaborately control the fusion of its disparate input sources. First, for encapsulating useful information from the pronunciation deviation patterns, we encode the phone confusion table (in a column-by-column manner) for each canonical phone into a continuous vector representation. Afterward, the corresponding attention vector $\mathbf{a}_n^{\mathrm{c}}$ for each degree vector $\mathbf{d}_n$ of the canonical phone q_n in the text prompt is calculated via multi-head attention:

$$\mathbf{a}_n^{\mathrm{c}} = MultiHead(\mathbf{d}_n, \mathrm{H}_n^{\mathrm{c}}), \tag{2}$$

where H_n^c is a matrix formed by splicing together the phone-level embedding of each element at the n-th column of the phone confusion matrix. For the multi-head attention operation, the query comes from $\mathbf{d}_n$ and the key-value pair comes from H_n^c, respectively. This way, we quantify the interactions between the embeddings of pronounced phones $H^p = (\mathbf{h}_1^p, \mathbf{h}_2^p, \dots, \mathbf{h}_L^p)$ and the matching degree matrix D as follows:

$$A^p = MultiHead(D, H^p). \quad (3)$$

The gate values are computed with a linear layer followed by a sigmoid function, whose input features include attention weight of pronunciation deviation patterns $A^c = (\mathbf{a}_1^c, \mathbf{a}_2^c, \dots, \mathbf{a}_N^c)$, the attention weight of possible pronounced phone A^p, and the phone-level embeddings of text prompt H^q. We denote the gate vectors for phonetic errors, (hypothesized) pronounced phonemes, and text prompts as $\mathbf{g}^c$, $\mathbf{g}^p$ and $\mathbf{g}^q$:

$$\mathbf{g}^c = \sigma(\mathbf{w}_c^T[A^c; D] + \mathbf{b}^c), \quad (4)$$

$$\mathbf{g}^p = \sigma(\mathbf{w}_p^T[A^p; D] + \mathbf{b}^p), \quad (5)$$

$$\mathbf{g}^q = \sigma(\mathbf{w}_q^T[H^q; D] + \mathbf{b}^q), \quad (6)$$

where $\mathbf{w}_c$, $\mathbf{w}_p$, $\mathbf{w}_q$, $\mathbf{b}^c$, $\mathbf{b}^d$ and $\mathbf{b}^q$ are learnable parameters, σ is the sigmoid function, and [·;·] denotes the concatenation of matrices. The fused representation $\mathbf{h}_n^f$ of the canonical phone q_n in the text prompt is derived through following equations:

$$\mathbf{d}_n^c = (W^{cg}\mathbf{d}_n + \mathbf{b}^{cg}) \cdot \mathbf{g}^c \quad (7)$$

$$\mathbf{d}_n^p = (W^{pg}\mathbf{d}_n + \mathbf{b}^{pg}) \cdot \mathbf{g}^p \quad (8)$$

$$\mathbf{d}_n^q = (W^{qg}\mathbf{d}_n + \mathbf{b}^{qg}) \cdot \mathbf{g}^q \quad (9)$$

$$\mathbf{h}_n^f = \mathbf{d}_n + \mathbf{d}_n^p + \lambda^c\mathbf{d}_n^c + \lambda^d\mathbf{d}_n^d \quad (10)$$

where "·" represents element-wise multiplication, λ^c and λ^d are tunable coefficients and their sum is set equal to 1; and W^{cg}, W^{pg}, W^{qg}, $\mathbf{b}^{cg}$, $\mathbf{b}^{pg}$, $\mathbf{b}^{qg}$ are parameters to be learned. Finally, the probability distribution of diagnostic feedback for the n-th canonical phone symbol is derived by

$$\boldsymbol{\theta}_{y_n} = softmax(W^d\mathbf{h}_n^f + b^d), \quad (11)$$

where W^d and b^d are learnable parameters.

2.8. Training Objectives

The learning of the proposed model is conducted in end-to-end manner, where the learning process is conducted by optimizing two objectives that correspond to mispronunciation detection and mispronunciation diagnosis, respectively:

$$\mathcal{L}_{det} = -\sum_{n=1}^{N} \log P_{det}(e_n|\mathbf{q}, X), \quad (12)$$

$$\mathcal{L}_{dia} = -\sum_{n=1}^{N} \log P_{dia}(y_n|\,\mathbf{q}, X), \quad (13)$$

where $\mathcal{L}_{det}$ is the negative log-likelihood for training the detector and $\mathcal{L}_{dia}$ is the negative log-likelihood for training the predictor. The two modules are linearly combined as the overall objective for model learning:

$$\mathcal{L} = \mathcal{L}_{det} + \beta\mathcal{L}_{dia}, \quad (14)$$

Table 1. Statistics of the experimental speech corpora.

	Corpus	Sets	Spks.	Utters.	Hrs.
L1	TIMIT	Train	462	3,696	3.14
		Dev.	50	400	0.34
		Test	24	192	0.16
L2	L2-ARCTIC	Train	15	2,249	2.29
		Dev.	3	400	0.49
		Test	6	900	0.87

where $\beta \in [0, 1]$ is a control parameter: a larger β value means a higher weight on the mispronunciation diagnosis subtask. mispronunciation detection is considered to be an easier subtask than mispronunciation diagnosis, because the former belongs to a binary classification problem while the latter is a multi-class classification problem. The value of β is set to be 0.3 based on the development set.

3. EXPERIMENTS

3.1. Speech Corpora

TIMIT [22] is a widely-used English read speech dataset that was designed for the development and evaluation of automatic speech recognition systems. TIMIT contains recordings of 630 speakers from 8 major dialects of American English.

L2-ARCTIC [23] is a recently released L2-English speech dataset, which was compiled for research on CAPT, accent conversion and others. L2-ARCTIC contains correctly pronounced utterances and mispronounced utterances of 24 non-native speakers (12 males and 12 females), whose mother-tongue languages include Hindi, Korean, Mandarin, Spanish, Arabic and Vietnamese.

3.2. Data Pre-processing

The transcripts of the two aforementioned speech corpora was converted to the corresponding canonical phone sequences with a set of 39 phones according to the definition of the CMU pronunciation dictionary. This phone set was additionally supplemented with an /unk/ phone to account for non-categorial or distortion-lkie pronunciation errors. In addition, the phone /ao/ and /zh/ were relabeled as /aa/ and /sh/, respectively, following the preprocessing recipes of Kaldi. Beyond this, we divided these two corpora into training, development and test sets, respectively. In particular, the setting of the mispronunciation detection and diagnosis experiments on L2-ARCTIC followed the recipe provided by [12]. Table 1 summarizes the detailed statistics of the TIMIT and L2-ARCTIC speech corpora. Furthermore, a data augmentation method was adopted to attenuate the overfitting problem. To this end, we randomly chose the correct pronunciation segments and replaces 30% of the phone symbols in the corresponding text prompt with other phone symbols to artificially generate more erroneous pronunciations for model training.

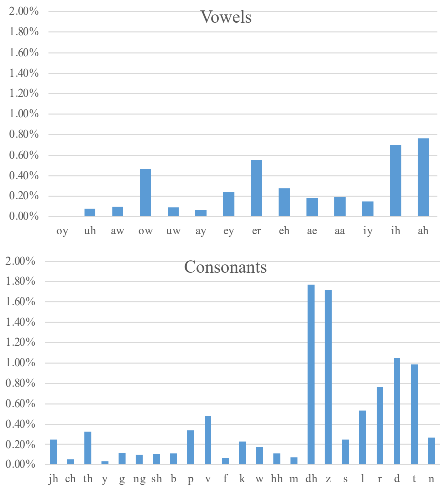

Fig. 3. A depiction of the pronunciation error patterns of the L2-ARCTIC test set.

3.3. Experimental Setting

We adopted a publicly-released pre-trained Transformer architecture[1] for initialization, which was trained with 960 hours of native English speech from the LibriSpeech dataset [25]. The pre-trained Transformer consists of a 12-layers encoder and 6-layers decoder with 8 attention heads of dimensions 512 and 2,048 units in the feed-forward layers. Our dictation module was initialized with a pretrained Transformer and then finetuned on TIMIT and L2-ARCTIC. The predictor was initialized with the decoder part of pretrained model, which consists of 6 Transformer blocks. The dimensionality of all the resulting embeddings and the output of hidden states is set to 512. The input to the dictation module is composed of 80-dimensional Mel-filter-bank feature vectors which are extracted every 10 ms with a Hanning window of size 25 ms. We trained our model with Adam optimizer and Noam learning rate decay as in [21], starting with a learning rate of 5.0, a warmup update step to 25,000, and a mini-batch size of 32. The parameters of the last 10 epochs were averaged and used for inference.

3.4. Baseline Methods

We will compare our method (viz. PeppaNet) against several methods to investigate the potential of our framework, as briefly described below: **Goodness of pronunciation** [8] is the most iconic pronunciation-scoring based method, which computes the ratio between the likelihoods of a canonical phone and the most likely pronounced phones predicted by a DNN-HMM-based speech recognition system. **CNN-RNN-CTC** [11] is a classical dictation-based neural model, which employs a stack of convolutional layers followed by a Bi-GRU layer and several fully-connected layers to generate a hypothesis for the pronounced phone sequence with the help of the CTC objective function. **Transformer** adopts an encoder-decoder attention model as a dictation model and instantiates the model structure with Transformer, whose model parameters were transferred from a large-scale L1 English speech recognition model and then finetuned on the two MDD corpora

[1]https://drive.google.com/file/d/17cOOSHHMKI82e1MXj4r2ig8gpGCRmG2p/view

Table 2. Experimental results of different methods on mispronunciation detection.

Models	Recall (%)	Precision (%)	F1-score (%)
GOP	69.97	32.54	44.42
CNN-RNN-CTC	74.19	34.70	47.28
Transformer	53.50	53.82	53.66
APL-2	54.49	52.79	53.62
Mask-CTC	51.71	52.32	52.01
PeppaNet	64.53	51.38	56.81

Table 3. Experimental results of different methods on mispronunciation diagnosis.

Models	FRR	FAR	Diagnosis Accuracy
CNN-RNN-CTC	24.45	24.81	55.89
Transformer	7.31	46.50	48.21
APL-2	8.88	45.51	66.06
PeppaNet	9.61	36.47	74.12

mentioned above. **APL-2** [26] employs a well-trained L1 English speech recognition model as an additional feature extractor to extract frame-level phonetic embeddings for building an attention-based dictation model. **Mask-CTC** merely utilizes the dictation module of our method, whose model parameters were transferred from a L1 English pretrained speech model and then finetune on the MDD corpora.

3.5. Quantitative Analysis

At the outset, we perform quantitative analysis on the phonetic distribution of pronunciation error patterns of the test set. With a closer look at Figure 3, we find that relatively large portions of the mispronunciations of vowels, like {/ah/, /ih/, /er/}, belong to the continuant phones, while a similar situation exists for consonants like {/dh/, /z/}. Furthermore, we can observe that the most frequently mispronounced consonants are {/dh/, /z/, /d/, /t/}. The articulation patterns of these consonants are characterized to have the anterior attribute, which is a peculiar property of the English language and thus is somewhat difficult for L2 learners to pronounce these consonants correctly.

3.6. Experimental Results

Table 2 presents the experimental results of all compared methods on mispronunciation detection. There are three noteworthy points to these results. First, it is evident that all the dictation-based methods outperform GOP by a significant margin in terms of the F1-score metric, which promises the potential of the dictation-based models. At another point, CNN-RNN-CTC achieves the highest recall value but has the downside of a poor precision value obtained.

Table 4. An ablation study on our proposed method.

Models	FRR	FAR
PeppaNet	9.61	36.47
− Phone Confusion Table	10.97	33.81
− Dictation Module	9.47	37.34
− Selective Gating Mechanism	12.20	47.24

Transformer and APL-2 are quite competitive with each other and have excellent mispronunciation detection performance on the L2-ARCTIC benchmark. Second, our proposed method performs the best in terms of F1-score, in relation to the other top-of-the-line methods. If we further compare our method with Mask-CTC (which uses merely the dictation module of our method), we confirm the benefit of integrating heterogeneous information cues and additional modules (*cf.* Section 2) for better mispronunciation detection, with respect to the recall and F1-score metrics. Going one step further, we report on the experimental results for mispronunciation diagnosis, which are shown in Table 3. We can observe that our method is far superior to all the methods compared in this paper in terms of diagnostic accuracy. This might be mainly attributed to the fact that our method adopts a unified model structure to integrate heterogeneous information cues derived from the alignment process and dictation process, while the training objective of our model directly optimizes both mispronunciation detection accuracy and diagnostic accuracy. Here we also list the FRR (False Rejection Rate) and FAR (False Alarm Rate) results in Table 3 for reference.

3.7. Ablation Study

To study the effectiveness of each module involved in our model structure, we conduct an ablation study with the following settings: 1) removing the dictation module, 2) removing the confusion table, and 3) replacing the selective gating mechanism with a simple aggregation strategy. Table 4 shows the corresponding results. First, it can be found that FRR increases when removing the phone confusion table, whereas FAR increase when removing the dictation module. These phenomena reveal that leveraging both the pronunciation deviation patterns and the information gathered form the dictation process contribute prominently to the design of MDD method. Next, when replacing the selective gating mechanism with a simple aggregation strategy, we obtain relatively worse performance with respect to the FRR and FAR metrics, thereby demonstrating the utility of the selective gating mechanism.

4. CONCLUSION AND FUTURE WORK

In this research, we have proposed a novel neural method (dubbed PeppaNet) for MDD, which builds on a unified structure to jointly model the dictation process and the alignment process. A series of empirical experiments conducted on the L2-ARCTIC benchmark dataset have revealed the feasibility of our proposed method in comparison to some top-of-the-line methods. As to future work, we plan to integrate more acoustic, phonetic and phonological cues into our model. We also envisage to explore sophisticated discriminative training objectives and effective pre-trained models.

5. ACKNOWLEDGEMENT

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